COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 11

PLANAR GRAPHS

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PLANAR GRAPHS

- A graph is called *planar* if it can be drawn in the plane without any edges crossing.
- Such a drawing is called a *planar representation* of the graph
- *Note:* A graph may be planar even if it is usually drawn with crossings, since it may be possible to draw it in another way without crossings.

Q : **Is K_4** (shown in Figure 2 with two edges crossing) **planar**?

Solution: **K_4 is planar** because it can be drawn without crossings, as shown in Figure 3

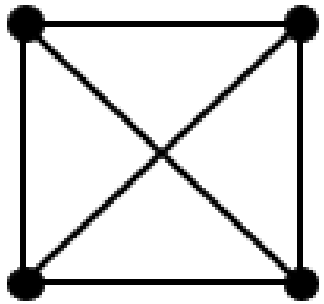


Figure 1

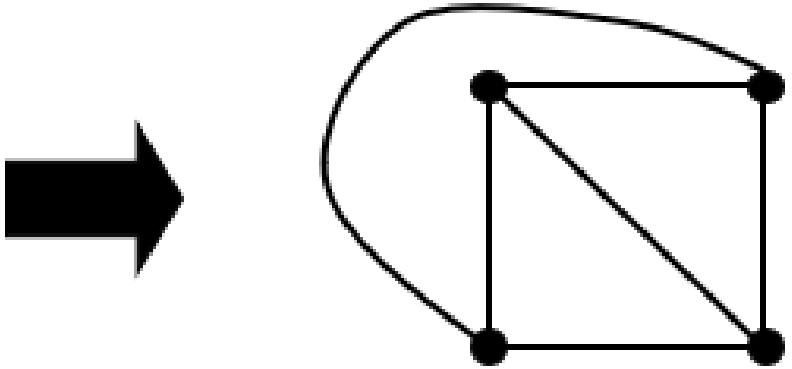


Figure 2

PLANAR GRAPHS

Q: **Is Q_3** (shown in Figure 4) **planar**?

Solution: **Q_3 is planar**, because it can be drawn without any edges crossing, shown in Figure 2.

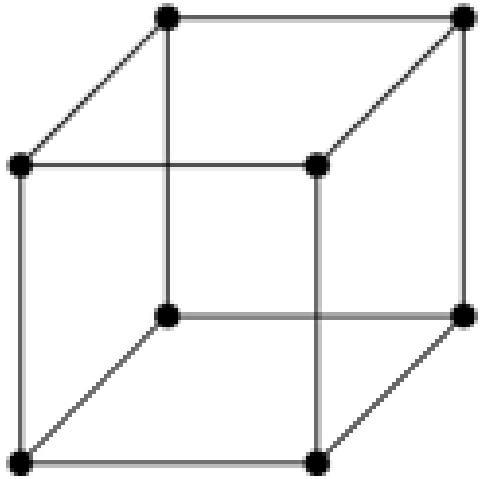


Figure 1

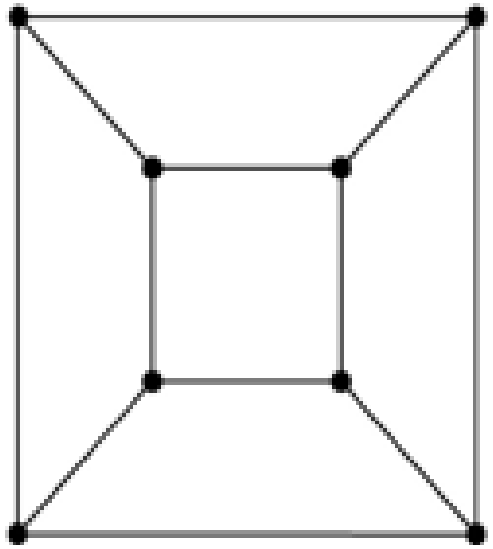


Figure 2

PLANAR GRAPHS

Q: Is $K_{3,3}$ (bipartite graph) shown in Figure 1, planar?

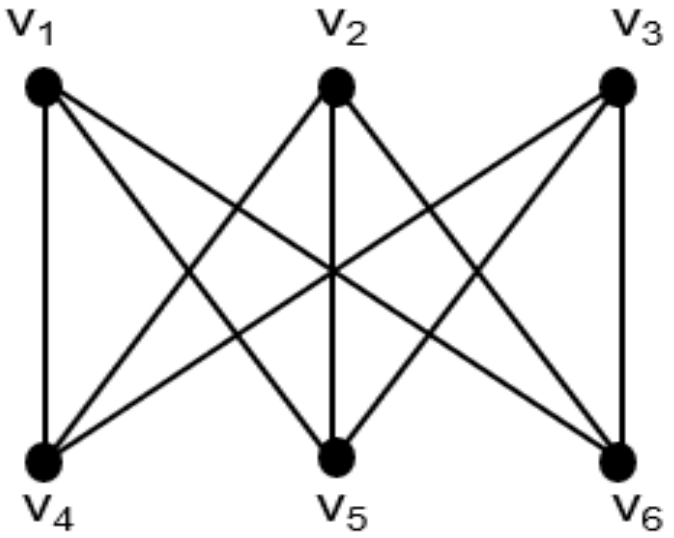


Figure 1

Solution: the graph $K_{3,3}$ must be nonplanar.

- In any planar representation of $K_{3,3}$, vertex v_1 must be connected to both v_4 and v_5, and v_2 also must be connected to both v_4 and v_5.
- Figure 2(a): the four edges $\{v_1, v_4\}$, $\{v_4, v_2\}$, $\{v_2, v_5\}$, $\{v_5, v_1\}$ form a closed curve that splits the plane into two regions, R_1 and R_2
- Figure 2(b): next, we note that v_3 must be in either R_1 or R_2. When v_3 is in R_2, then the edges $\{v_3, v_4\}$ and $\{v_3, v_5\}$ separate R_2 into two sub-regions, R_{21} and R_{22}.

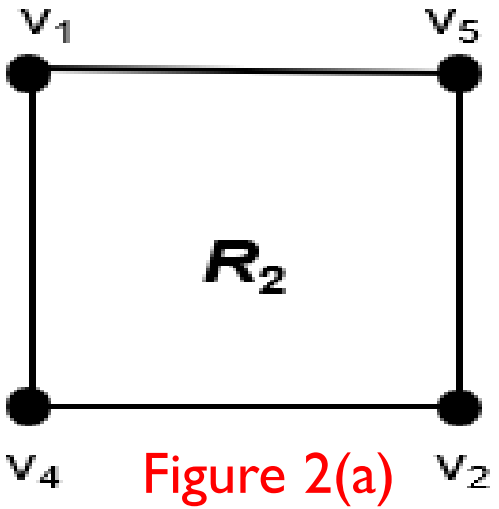


Figure 2(a)

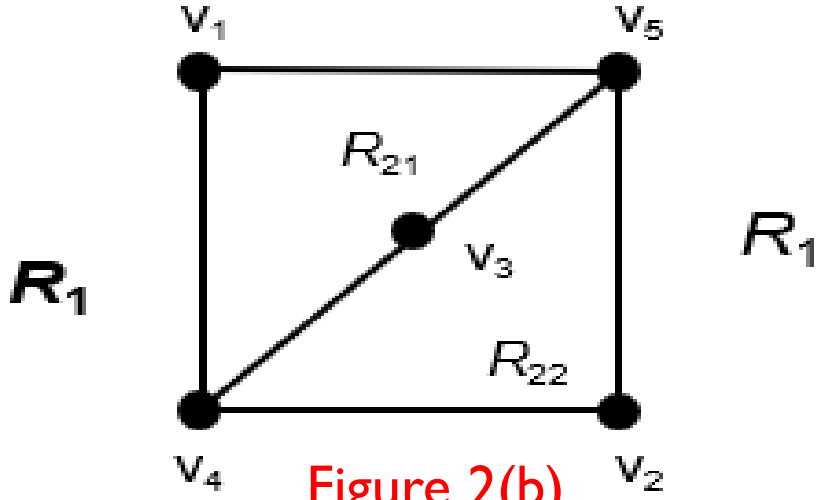


Figure 2(b)

PLANAR GRAPHS

- Now there is no way to place vertex v_6 without forcing a crossing:
  - If v_6 is in R_1, then $\{v_6, v_3\}$ must cross an edge
  - If v_6 is in R_{21}, then $\{v_6, v_2\}$ must cross an edge
  - If v_6 is in R_{22}, then $\{v_6, v_1\}$ must cross an edge

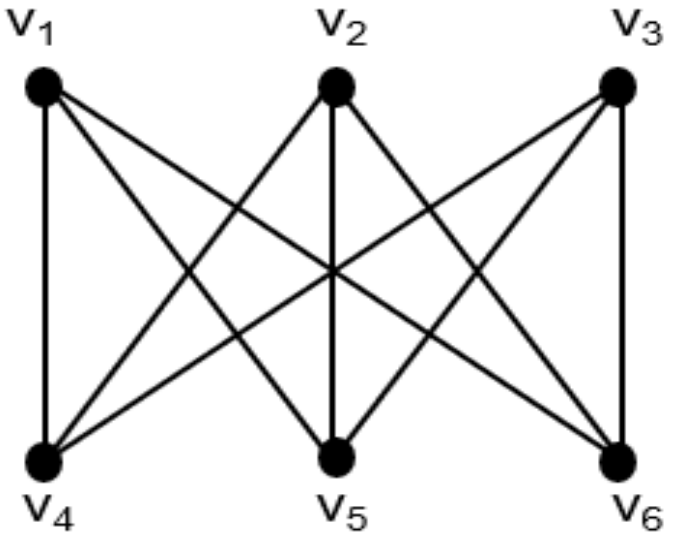


Figure 1

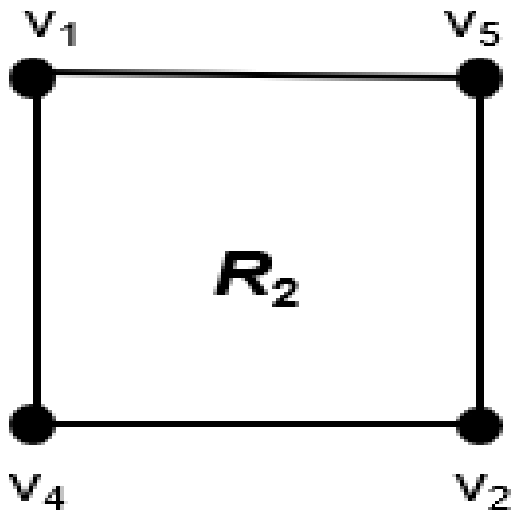


Figure 2(a)

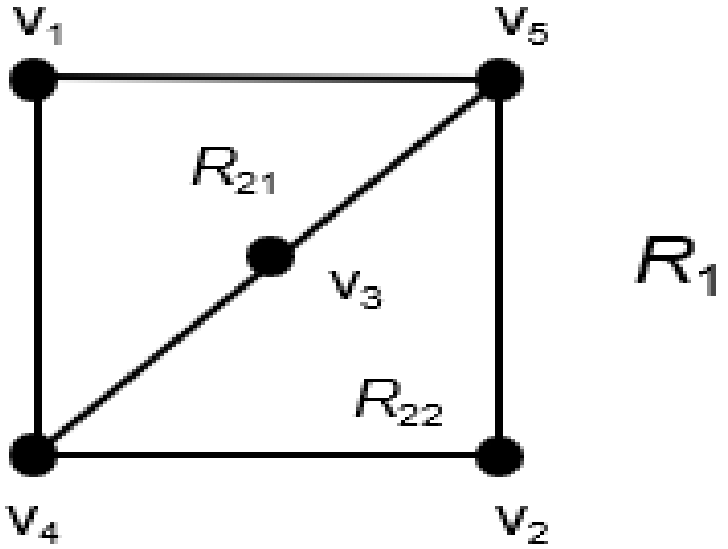


Figure 2(b)

PLANAR GRAPHS

- **When v_3 is in R_1**, then the edges $\{v_3, v_4\}$ and $\{v_4, v_5\}$ separate R_1 into two sub-regions, R_{11} and R_{12}.
- Now there is no way to place vertex v_6 without forcing a crossing:
  - If v_6 is in R_2, then $\{v_6, v_3\}$ must cross an edge
  - If v_6 is in R_{11}, then $\{v_6, v_2\}$ must cross an edge
  - If v_6 is in R_{12}, then $\{v_6, v_1\}$ must cross an edge

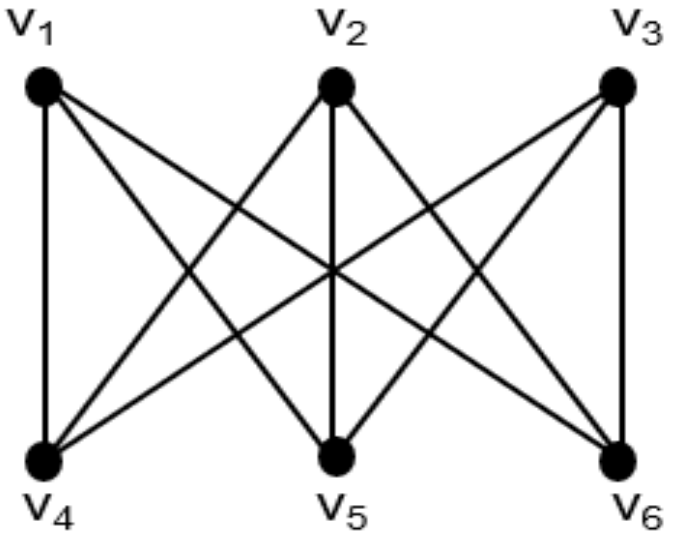


Figure 1

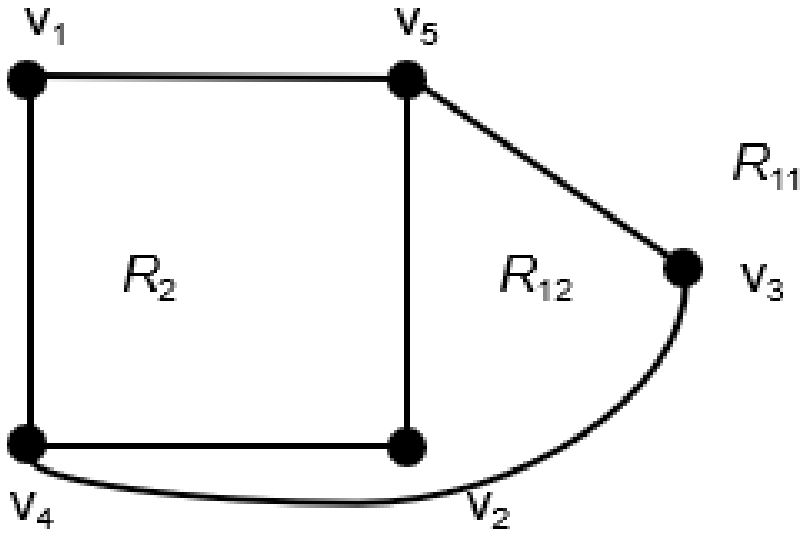


Figure 2(c)

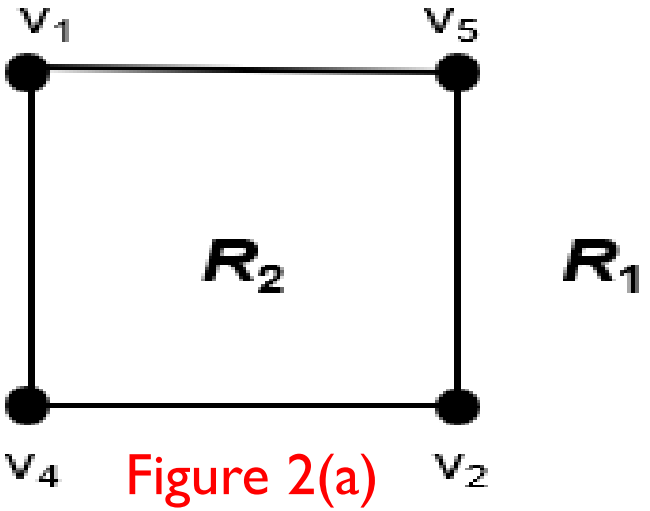


Figure 2(a)

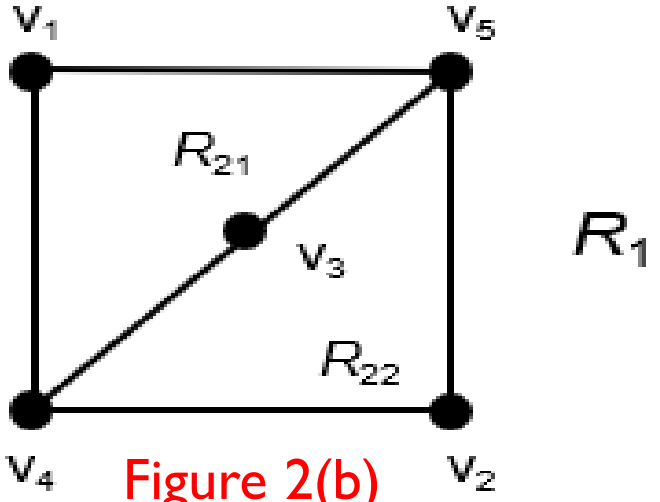
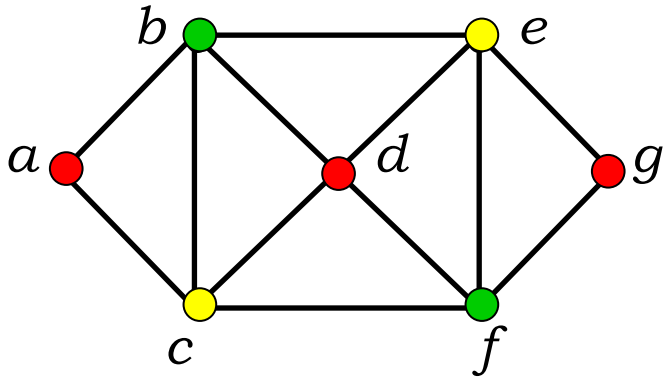


Figure 2(b)

GRAPH COLORING

- A ***coloring*** of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.
- The ***chromatic number*** of a graph is the least number of colors needed for a coloring of this graph.
- *Theorem:* The chromatic number of a planar graph is no greater than four.

Q. What is the chromatic number of the graph shown below?

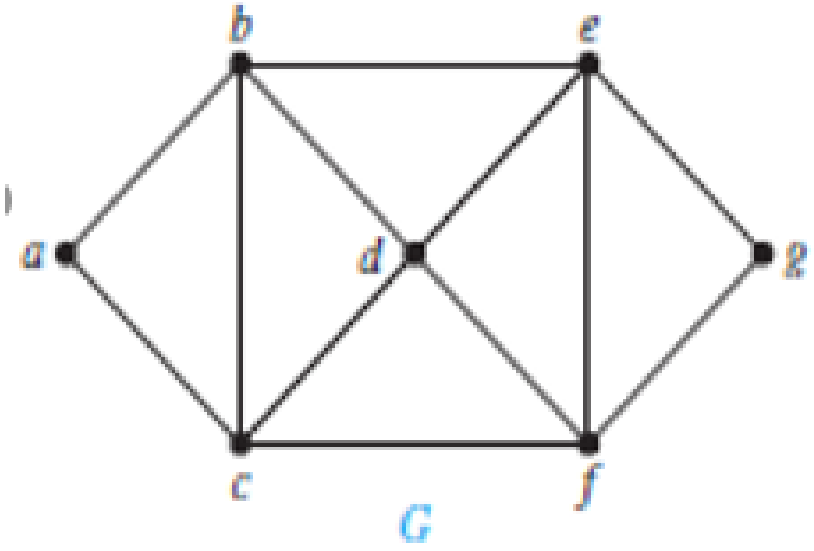


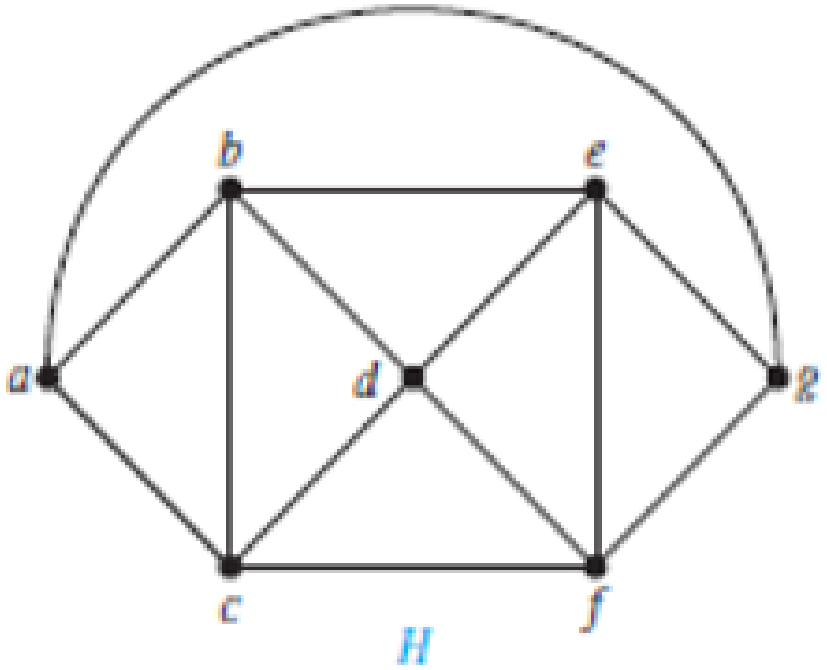
Solution:

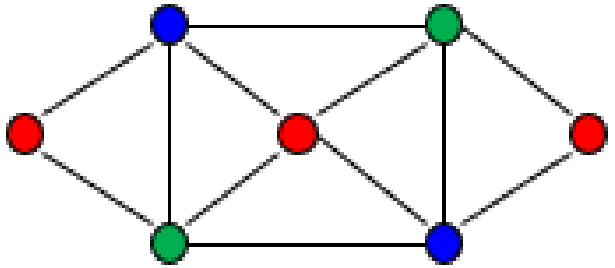
- The chromatic number must be at least 3 since *a*, *b*, and *c* must be assigned different colors.
- So Let's try 3 colors first. 3 colors work, so the chromatic number of this graph is 3.

GRAPH COLORING

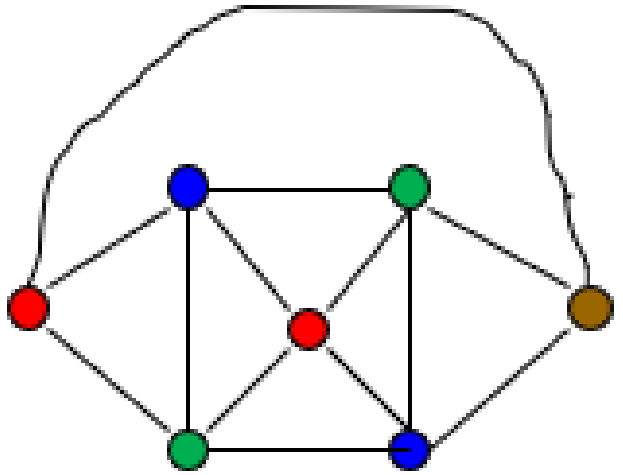
Q. What are the chromatic numbers of the graphs G and H?







G



H

Solution: Chromatic number of (*G*) = 3
Chromatic number of (*H*) = 4

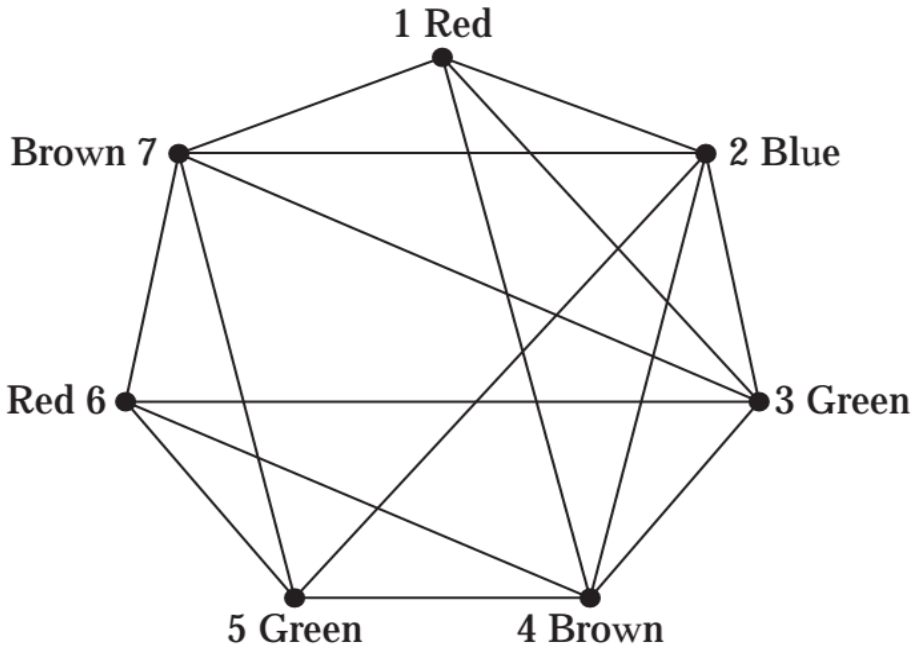
AN APPLICATION OF GRAPH COLORING

Q: How can the final exams at a university be scheduled so that no student has two exams at the same time?

Solution: This scheduling problem can be solved using a graph model, with vertices representing courses and with an edge between two vertices if there is a common student in the courses they represent. Each time slot for a final exam is represented by a different color. A scheduling of the exams corresponds to a coloring of the associated graph.

- Suppose there are seven finals to be scheduled. Suppose that the following pairs of courses have common students: 1 and 2, 1 and 3, 1 and 4, 1 and 7, 2 and 3, 2 and 4, 2 and 5, 2 and 7, 3 and 4, 3 and 6, 3 and 7, 4 and 5, 4 and 6, 5 and 6, 5 and 7, and 6 and 7.

Time Period	Courses
I	1, 6
II	2
III	3, 5
IV	4, 7



REFERENCES

- *Discrete Mathematics and its applications with combinatorics and graph theory (7th edition)* by Kenneth H. Rosen [Indian Adaptation by KAMALA KRITHIVASAN], published by McGraw-Hill, is the required text.
- *SCHAUM'S ouTlines Discrete Mathematics(2nd edition)*, by Seymour Lipschutz, Marc Lipson
- *Discrete and Combinatorial Mathematics(3rd Edition)* by *R. P. Grimaldi,*, published by Addison-Wesley